

TAMIL NADU STATE BOARD - STANDARD 11 PHYSICS VOLUME 2

CHAPTER 11: WAVES

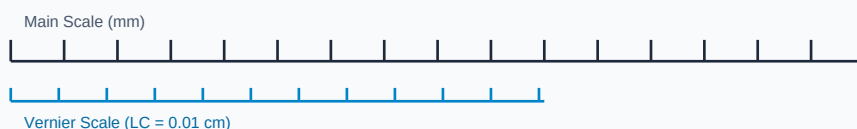
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 11 English Medium
Subject:	Physics Volume 2
Chapter Name:	Waves

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Measurement Linear Scale Division

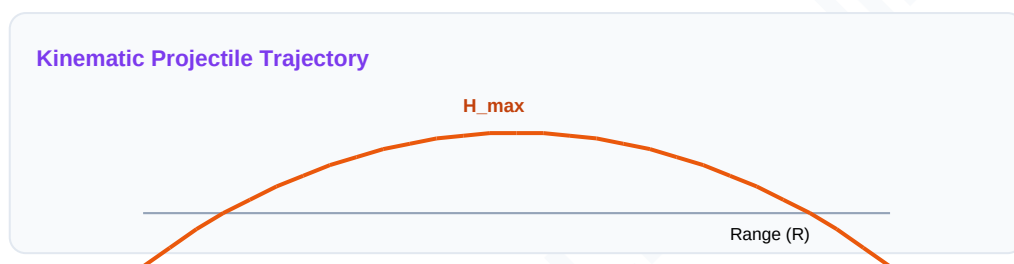


Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Transverse and Longitudinal Waves	Transverse waves oscillate perpendicular to propagation (e.g. light); longitudinal waves oscillate parallel (e.g. sound).
Beats	The periodic variation in sound intensity heard when two waves of slightly different frequencies interfere.
Doppler Effect	The apparent change in frequency of a wave due to relative motion between the source and observer.

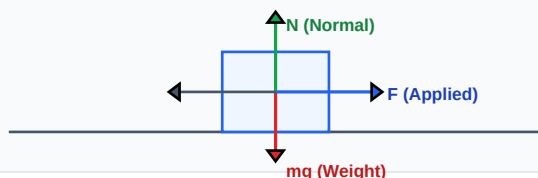
Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Force Vector Free Body Diagram (FBD)



Essential Formulas, Laws & Identities:

- Wave Velocity: $v = f * \lambda$ | Speed of Sound (gas): $v = \sqrt{\gamma * P / \rho}$
- Apparent Doppler Frequency: $f' = f * [(v \pm v_o) / (v \mp v_s)]$
- Standing Wave (string fixed at ends): $f_n = n * v / (2 * L)$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

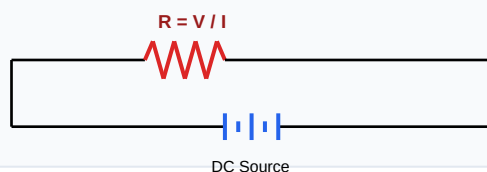
Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: A train blowing a whistle of frequency 400Hz approaches a station at 30 m/s. Find apparent frequency heard by listener on platform. (Speed of sound = 330 m/s)

Step-by-step Solution:

$$f' = f * [v / (v - v_s)] = 400 * [330 / (330 - 30)] = 400 * [330 / 300] = 440 \text{ Hz.}$$

Schematic Electric Circuit Diagram



HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Explain Newton's formula for the velocity of sound in air and Laplace's correction to it.

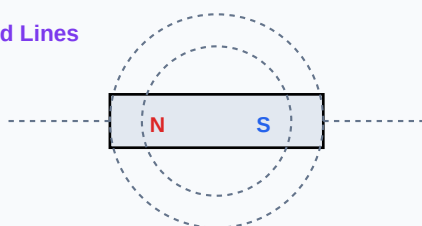
Analytical Solution Strategy:

Newton assumed isothermal conditions: $v = \sqrt{P/\rho} = \sqrt{1.013 \times 10^5 / 1.29} = 280 \text{ m/s}$ (underestimates).

Laplace corrected that compression is adiabatic: $v = \sqrt{\gamma P/\rho}$. For air $\gamma=1.4$.

$v = \sqrt{1.4 \times 280^2} = 331 \text{ m/s}$ (matches experimental values).

Magnetic Dipole Bar Field Lines



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): State the principle of superposition of waves.

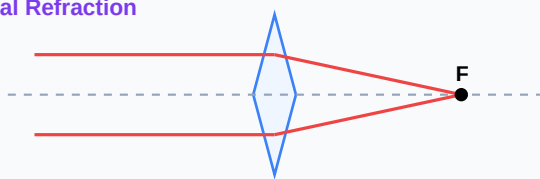
Q2 (2-Mark): Explain how beats are formed and state the formula for beat frequency.

Q3 (2-Mark): Show that only odd harmonics are produced in a closed organ pipe.

Q4 (5-Mark): A wave is given by $y = 5\sin(100t - 2x)$. Find frequency and wavelength.

Q5 (5-Mark): List three applications of the Doppler effect in modern technology.

Convex Lens Optical Refraction



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Medical Imaging

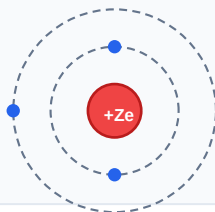
Ultrasound scanners check blood velocity maps using Doppler shifts of reflections.

Application Case: Astro Navigation

Radar speed guns measure automobile speeds using radio frequency shifts.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Bohr Atomic Shell Configuration



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Sinusoidal AC Voltage Waveform

